data. It is the basis of the Garnet Framework (https://garnet-framework.tech/), an open-source framework for building digital twins that use knowledge graphs.

5.4.3 Graph traversals

When modeling the real world in a knowledge graph, traversal is how we discover insights and answer complex questions. *Traversing* a graph involves identifying a starting point—a specific node—and then following the relationships to other nodes recursively until you find the information you are looking for, often using efficient graph traversal algorithms such as breadth-first, or depth-first search. Taking the example of the small subset of the knowledge graph representing a hotel shown in figure 5.6, I can build and traverse the graph using the declarative, open query language Cypher. Listing 5.3 shows how I use Cypher to first build the graph in a Memgraph server (running locally in a Docker container), and then run a traversal against it to find the battery level of all equipment on level two of the hotel.

NOTE

Memgraph (https://memgraph.com/) is an in-memory graph database designed for real-time analytics and transactional workloads on highly connected data.

Before running the code, start Memgraph in a Docker container with the following command:

```
docker run -p 7687:7687 -p 7444:7444 --name memgraph memgraph/memgrap
h-mage
```